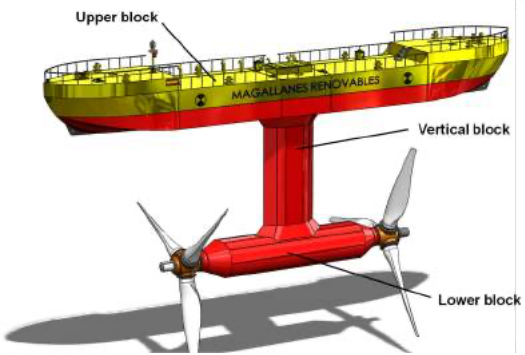

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MAGALANES PROJECT
Mooring system design of a floating Tidal energy converter (Summary)

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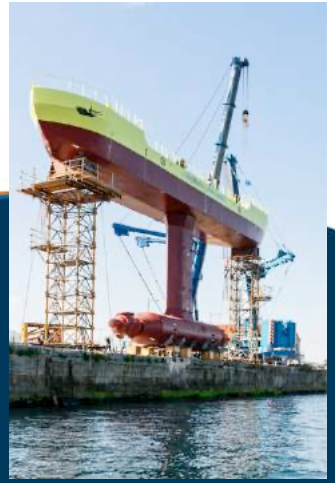
SYSTEM CERTIFICATION ISO 9001 SGS

Introduction - ATIR platform

- The Magallanes Tidal Energy Converter (TEC) is an offshore floating tidal energy platform, named ATIR, which will be deployed at the EMEC tidal testing site in the Fall of Warness (FoW), Scotland at Berth 1, in a water depth of 49 meters (LAT). The Magallanes platform will be carrying two tidal turbines with a combined rated power output of 1.7MW.

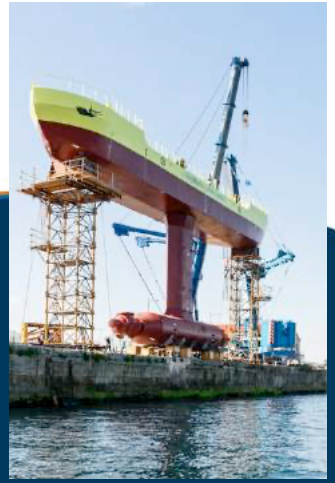
- The design of this TEC and engineering solutions were developed by Sea Master C&E.

- This work describes shortly the mooring system design and engineering solutions for the ATIR-Offshore platform (Magallanes Renovables), under certain environmental conditions.



Introduction - ATIR platform

- It will be in place for a period of at least 12 months to give data and validation for year-round events. The mooring system consists of four lines and four rock anchors (one per line) (two forward and two rear) which has been designed for water-depth ranging from 40 to 50 meters. It has been adapted to survival conditions in 10-year Return Conditions, according the Bureau Veritas, Classification of Mooring Systems for Permanent and Mobile Offshore Units (NR 493 DT R03, Dec. 2015) and DNV GL Offshore Standard, Position Mooring, DNV-OS-E301, July 2015).



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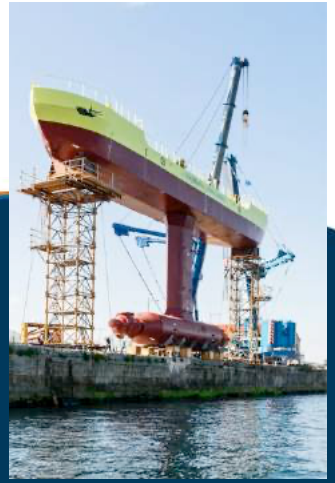
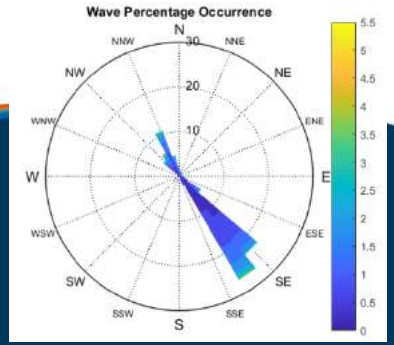


Environmental conditions - Waves

- Meteocean data (wave, wind and tide) for the Fall of Warness was provided by European Marine Energy Centre Ltd. (EMEC).

- Waves predictions based on scatter diagram data to joint probability of significant wave height and peak wave period

AVERAGE PERCENTAGE FREQUENCY OCCURRENCE OF Hs (ANNUAL) in %											
Hs[m]/Tp[s]	0.5	1	1.5	2	2.5	3	3.5	4	4.5	>4.5	
0-4	1.13	3.88	2.83	1.63	0.62	0.14	0.00	0.00	0.00	0.00	10.23
4-6	6.08	6.80	4.44	3.17	2.35	1.45	0.73	0.34	0.11	0.03	25.30
6-8	11.84	9.44	3.53	1.59	0.84	0.51	0.24	0.17	0.05	0.03	28.23
8-10	5.79	5.43	2.91	1.05	0.32	0.09	0.04	0.02	0.01	0.01	15.68
10-12	3.21	3.10	1.83	0.95	0.28	0.12	0.03	0.01	0.00	0.00	9.53
12-14	1.80	2.30	1.30	0.75	0.28	0.08	0.02	0.01	0.00	0.00	6.53
14-16	0.89	1.47	1.00	0.56	0.14	0.04	0.00	0.00	0.00	0.00	4.10
>16	0.12	0.12	0.09	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.40
Total	30.86	32.35	17.94	9.77	4.82	2.42	1.06	0.54	0.17	0.07	100

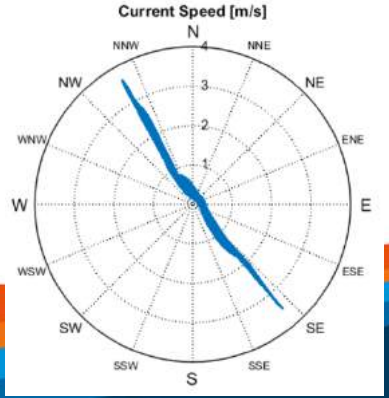


Wave percentage occurrence (wave height in m)

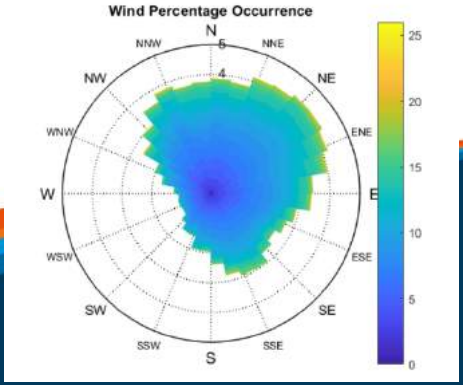
Environmental conditions - Wind & Current

- Wind (less directional dependence in the wind conditions than the wave conditions, with extreme conditions exceeding 20m/s occurring for a wide range of directions).

- Current speed (for current speeds greater than 1m/s, there is only a small variation in the current direction for a given velocity and the currents can be treated as effectively bi-directional).



Current speed (direction TO) for each direction



Wind percentage occurrence (wind speed in m/s)

Analysis Process - Summary

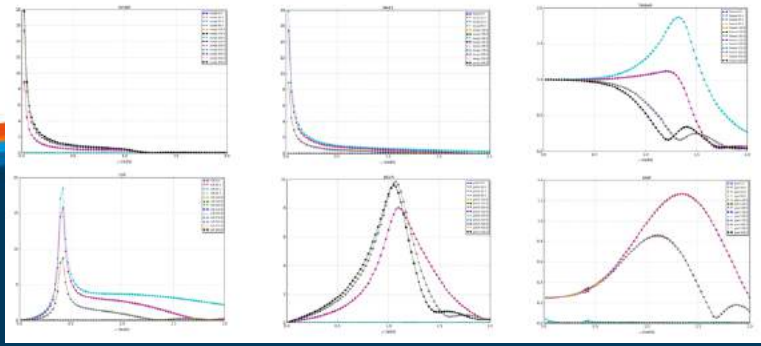
- For the evaluation of the loads on the mooring lines a quasi-dynamic approach has been followed. The quasi-dynamic analysis of mooring system is based in the time domain simulation of the mooring system response in specified environmental conditions (wind, current and drift forces produced by sea waves are compensated by the reactions in the mooring lines).

- A quasi-dynamic analysis procedure consists of the determination of the low frequency response of the moored object under the effect of waves, wind and current, followed by superimposition of the **wave frequency motions**.

Analysis Process - Summary

- An ARIANE (BV) model will be created. The diffraction model will be developed within HydroSTAR (BV). The current coefficients will be based on empirical tests reported here, and backed up by codes.

- In this section the analysis procedures to be followed within the design process are explained. For this study, seven different tasks will be established:
    - Hydrodynamics analysis (including estimation of drag currents coefficients)
    - Quasi-dynamic analysis
    - Maximum loads on hull attachment point
    - Max. and Mean loads on Anchors
    - Maximum Offsets or excursion
    - Analysis of cable connection

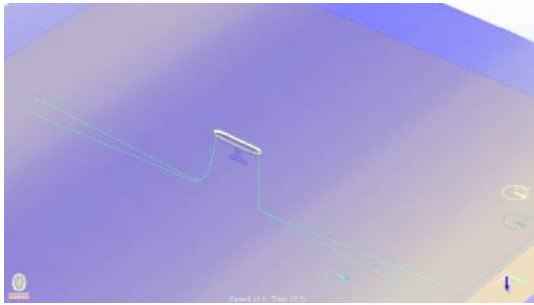


DISPLACEMENT RAOs of ATIR platform (HydroSTAR))

Mooring system design - Overview

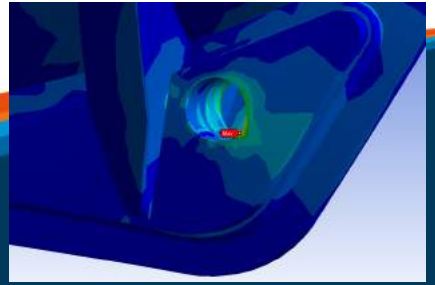
- The mooring system is a 4-legged catenary system made up of four chains and will be anchored to the seabed using four rock anchors (or deadweights) with a chain clump weight of 30t in each line).

- Following assessment of 4-mooring legs is a spread mooring which the angle increased to 6 degrees between 2 fore-lines and between 2 aft-lines. This mooring system design consists of 4 chain mooring lines of 225m length and a chain diameter of 76mm. The horizontal distance between the anchorage point and the mooring point on the platform will be 175m.



Layout of 4-mooring legs (ARIANE)

Underwater detail of mooring connection point



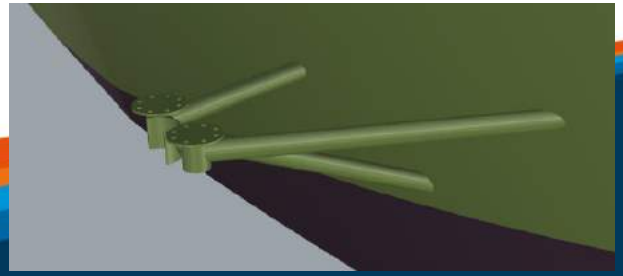
FEM simulation of mooring connection point

Umbilical Connection - Cable

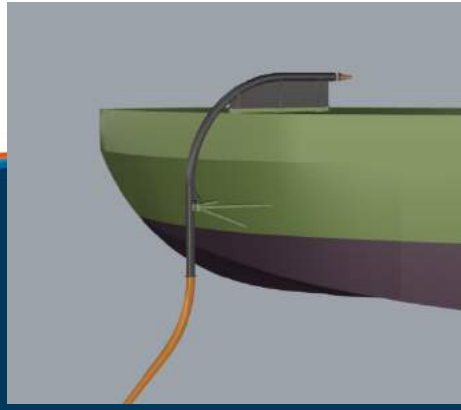
- The fundamental aim is then to design a system such that the following criteria are met:
  - Provide protection for the cable at the device interface
  - Ensure the minimum bend radius of the cable is not exceeded during maximum excursions;
  - Ensure the safe allowed tension of the cable is not exceeded during maximum excursions;
  - To with stand one year of testing.

- Connection of the dynamic cable to the ATIR Platform (see pictures below)

Device Interface



Tidal Device attachment point



Tidal Device attachment point

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Thank you

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